The Condensed GAGA Conjecture: A Proof-Theoretic Pre-Registration for the Clausen-Scholze Analytic Geometry Program

Paper 107 of the Constructive Reverse Mathematics Series

First CRM Prediction Paper

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Abstract

We formulate the first *predictive* conjecture of the Constructive Reverse Mathematics (CRM) program. The Archimedean Obstruction Thesis (AOT; Paper 98, Theorem 5.1) asserts that classical logic enters arithmetic geometry exclusively through Betti realization — infinite-dimensional topology over the archimedean continuum. Paper 106 identified GAGA (Géométrie Algébrique et Géométrie Analytique) descent (Montel's theorem $\Rightarrow$ Cartan A/B) as the unique CLASS component of the regular holonomic Riemann-Hilbert correspondence. We conjecture that the Clausen-Scholze condensed reformulation of Serre's GAGA theorem eliminates Montel's theorem, replacing topological compactness with the homological exactness of liquid modules over profinite sets, reducing the CRM cost from CLASS to at most $\mathsf{BISH} + \mathsf{WKL} + \mathsf{WLPO} = \mathsf{WKL} \vee \mathsf{WLPO}$, which is strictly weaker than LPO.

Three formal claims: (A) **The Condensed GAGA Conjecture:** the condensed reformulation eliminates Montel's theorem, reducing the CRM cost of GAGA descent from CLASS to at most $\mathsf{WKL} \vee \mathsf{WLPO} < \mathsf{LPO}$; (B) **The Riemann-Hilbert Descent:** if the conjecture holds, the overall CRM cost of the categorical equivalence drops from CLASS to LPO, isolating Deligne's canonical extension (the floor function on $\mathbb{R}$) as the unique maximum obstruction; (C) **The Falsifiability Wager:** a future CRM audit of Clausen-Scholze's completed proof will strictly confirm or refute the Archimedean Obstruction Thesis. The Lean 4 formalization (2 files, 0 `sorry`) implements the conjecture as an axiomatic API contract: the type-checker verifies that discharging the condensed axiom at the claimed CRM level produces the predicted phase transition.

1 Introduction

The first 106 papers of this series have been *descriptive*: given an existing proof, decompose it into atomic components and classify each by its CRM logical cost. This paper marks a transition. We use the structural patterns accumulated across 100+ audits to make a *predictive* claim about mathematics that has not yet been completed.

The prediction is sharp and falsifiable. It concerns the Clausen-Scholze analytic geometry program [5], which proposes to replace classical complex-analytic methods (Fréchet spaces, Montel's theorem, Cartan's theorems) with purely algebraic constructions using condensed mathematics and liquid modules. The Liquid Tensor Experiment [6] formalized a foundational input to this program in Lean, establishing that certain homological computations in the condensed setting are finitary.

Our claim: if Clausen-Scholze complete the condensed reformulation of Serre's GAGA theorem [17], they will inadvertently prove that the CLASS obstruction identified in Paper 106 is an *eliminable artifact* of 20th-century point-set topology, not a structural requirement of complex algebraic geometry.

1.1 Main results

Conjecture 1.1 (Conjecture A — The Condensed GAGA Deflation)**.** *Let X be a smooth projective complex algebraic variety. The Clausen-Scholze condensed reformulation of Serre's GAGA theorem eliminates the reliance on Montel's theorem and Fréchet sequential compactness, replacing them with the homological exactness of liquid modules over profinite sets. Consequently, the CRM logical cost of GAGA descent drops strictly from CLASS to at most WKL $\vee$ WLPO, which is strictly weaker than LPO.*

Corollary 1.2 (Corollary B — The Riemann-Hilbert Descent)**.** *Assume Conjecture 1.1. Then the overall CRM cost of the regular holonomic Riemann-Hilbert correspondence (Paper 106) strictly descends from CLASS to LPO. The archimedean continuous topology is eliminated as an obstruction, leaving Deligne's canonical extension — which requires real trichotomy via the floor function $\lfloor\cdot\rfloor : \mathbb{R} \to \mathbb{Z}$ (Bridges-Richman [2], equivalent to LPO) — as the sole non-constructive bottleneck.*

Theorem 1.3 (Theorem C — The Falsifiability Wager)**.** *The Archimedean Obstruction Thesis (Paper 98, Theorem 5.1) is strictly falsifiable at the Riemann-Hilbert correspondence:*

- *If Clausen-Scholze's condensed GAGA requires CLASS (e.g., proving liquid nuclearity internally requires topological compactness over $\mathbb{R}$), the AOT is fundamentally flawed.*
- *If the cost drops to WKL + WLPO or below, the AOT is confirmed: classical logic in the RH correspondence is exclusively an artifact of archimedean topology.*

1.2 CRM primer

We work in Bishop's constructive mathematics (BISH) as the base system. The hierarchy of principles relevant to this paper is:

$$\mathsf{BISH} \subset \mathsf{BISH} + \mathsf{LLPO} \subset \mathsf{BISH} + \mathsf{LPO} \subset \mathsf{CLASS}.$$

WKL (Weak König's Lemma) and WLPO (Weak Limited Principle of Omniscience) sit strictly between LLPO and LPO in the lattice. Crucially, WKL and WLPO are *incomparable*: neither implies the other over BISH (Ishihara [9]). Their join WKL $\vee$ WLPO is *strictly weaker than* LPO, since LPO $\Leftrightarrow$ WLPO + MP (Bridges-Richman [2]) and WKL does not imply MP (Markov's Principle: the unbounded search principle). LPO asserts that every binary sequence has a 1 or is all zeros. See Paper 1 [13] for full definitions.

1.3 State of the art

Clausen and Scholze introduced condensed mathematics in [4], replacing topological spaces with sheaves on the site of profinite sets. The analytic geometry program [5] extends this to replace Fréchet spaces with liquid modules — a specific class of condensed $\mathbb{C}$-vector spaces with controlled completeness. The Liquid Tensor Experiment [6] formalized the foundational exactness result in Lean, proving that certain Ext-groups vanish for liquid modules. None of these works address the constructive status of their arguments; the CRM perspective is new.

Paper 106 [16] executed the first CRM audit of the Riemann-Hilbert correspondence, yielding 10 BISH + 4 WLPO + 1 LPO + 1 CLASS = 16 components. The unique CLASS component is GAGA descent. The present paper tests the AOT (Paper 98 [14]) by predicting its eliminability.

1.4 Atlas fit

This paper extends the CRM pipeline from description to prediction. It complements Paper 106 (RH audit), Paper 98 (AOT statement), and Paper 99 (AOT validation via Hecke theta). The condensed GAGA conjecture is the natural boundary test for the AOT: if the thesis survives at the exact interface between algebraic and analytic geometry, it survives everywhere the CRM program has reached.

2 Preliminaries

Definition 2.1 (Omniscience principles [2])**.** LPO: $\forall \alpha : \mathbb{N} \to \{0,1\}$, either $\exists n.\, \alpha(n) = 1$ or $\forall n.\, \alpha(n) = 0$.
WLPO: $\forall \alpha : \mathbb{N} \to \{0,1\}$, either $\forall n.\, \alpha(n) = 0$ or $\neg(\forall n.\, \alpha(n) = 0)$.
LLPO: $\forall \alpha : \mathbb{N} \to \{0,1\}$ with at most one 1, either all even-indexed terms are 0 or all odd-indexed terms are 0.

Definition 2.2 (Weak König's Lemma [18])**.** WKL: every infinite *computably bounded* binary tree (i.e., every infinite subtree of $\{0,1\}^{<\mathbb{N}}$ with computable branching bound) has an infinite path. Equivalently (over RCA_0): the Maximum Theorem (every continuous function $f : [0,1] \to \mathbb{R}$ attains its supremum) or the Heine-Borel covering theorem for $[0,1]$ (Simpson [18], Theorems IV.1.2 and IV.2.3).

Definition 2.3 (The CRM partial order)**.** The principles relevant to this paper form a *partial order*, not a total order:

$$\begin{tikzcd}
 & \mathsf{CLASS} \arrow[d, dash] & \\
 & \mathsf{LPO} \arrow[d, dash] & \\
 & \mathsf{WKL} \vee \mathsf{WLPO} \arrow[dl, dash] \arrow[dr, dash] & \\
\mathsf{WKL} \arrow[dr, dash] & & \mathsf{WLPO} \arrow[dl, dash] \\
 & \mathsf{LLPO} \arrow[d, dash] & \\
 & \mathsf{BISH} &
\end{tikzcd}$$

Key relations: $\mathsf{BISH} \leq \mathsf{LLPO} \leq \mathsf{WKL} \leq \mathsf{LPO} \leq \mathsf{CLASS}$ and $\mathsf{BISH} \leq \mathsf{LLPO} \leq \mathsf{WLPO} \leq \mathsf{LPO} \leq \mathsf{CLASS}$, but $\mathsf{WKL} \not\leq \mathsf{WLPO}$ and $\mathsf{WLPO} \not\leq \mathsf{WKL}$ (Ishihara [9]).

The join $\mathsf{WKL} \vee \mathsf{WLPO}$ is *strictly weaker than* LPO. Since $\mathsf{LPO} \Leftrightarrow \mathsf{WLPO} + \mathsf{MP}$ (Bridges-Richman [2]) and WKL does not imply Markov's Principle MP (the unbounded search principle: $\neg\forall n.\, \alpha(n) = 0 \;\Rightarrow\; \exists n.\, \alpha(n) = 1$), the system $\mathsf{WKL} + \mathsf{WLPO}$ lacks the search component that upgrades WLPO to LPO.

Definition 2.4 (Serre's GAGA theorem [17])**.** Let X be a projective complex algebraic variety. The analytification functor $(\cdot)^{\mathrm{an}} : \mathrm{Coh}(X) \to \mathrm{Coh}(X^{\mathrm{an}})$ is an equivalence of categories. The classical proof requires Cartan's Theorems A and B: finite-dimensionality of coherent sheaf cohomology on compact complex spaces.

Definition 2.5 (Condensed mathematics [4])**.** A *condensed set* is a sheaf on the pro-étale site of the point, equivalently a sheaf on the category of profinite sets with jointly surjective finite families as covers. A *condensed* $\mathbb{C}$*-module* is a sheaf of $\mathbb{C}$-vector spaces on this site. A *liquid* $\mathbb{C}$*-vector space* is a condensed $\mathbb{C}$-module satisfying an exactness condition: the Breen-Deligne resolution admits explicit norm bounds ensuring vanishing of higher Ext-groups [5].

Definition 2.6 (Nuclearity [5])**.** A liquid $\mathbb{C}$-vector space V is *nuclear* if $V \otimes_{\mathbb{C}}^{\mathrm{liq}} W \xrightarrow{\sim} \mathrm{Hom}_{\mathbb{C}}^{\mathrm{liq}}(V^\vee, W)$ for all liquid W. Equivalently, V admits a Breen-Deligne resolution: a bounded complex of direct sums of $\mathbb{C}[S_i]^{\mathrm{liq}}$ for profinite S_i.

Definition 2.7 (Montel's theorem)**.** Every bounded sequence of holomorphic functions on a domain $U \subseteq \mathbb{C}^n$ admits a subsequence converging locally uniformly on U.

No proofs appear in this section.

3 The Classical Obstruction: Anatomy of Montel's Theorem

We trace the exact logical path from the RH correspondence to its CLASS obstruction.

3.1 From GAGA to Cartan A/B

Serre's GAGA theorem [17] establishes that for a projective complex variety X, the analytification functor is an equivalence $\mathrm{Coh}(X) \xrightarrow{\sim} \mathrm{Coh}(X^{\mathrm{an}})$. Essential surjectivity requires showing that every coherent analytic sheaf on X^{an} is algebraizable. Serre reduces this to the finite-dimensionality of $H^q(X^{\mathrm{an}}, \mathcal{F})$, which he establishes via Cartan's Theorems A and B [3].

3.2 From Cartan A/B to Schwartz's lemma

Cartan's theorems establish that on a Stein compact K, the restriction homomorphism

$$\Gamma(U, \mathcal{F}) \to \Gamma(K, \mathcal{F})$$

has finite-dimensional kernel and cokernel. The proof models $\Gamma(K, \mathcal{F})$ as a Fréchet space (locally convex, metrizable, complete) and applies Schwartz's lemma: a completely continuous (compact) linear operator between Fréchet spaces has a finite-dimensional kernel.

3.3 From Schwartz to Montel

To establish that the restriction operator $\Gamma(U_1, \mathcal{O}) \to \Gamma(U_2, \mathcal{O})$ between nested Stein compacta is compact, the proof invokes Montel's theorem (Definition 2.7).

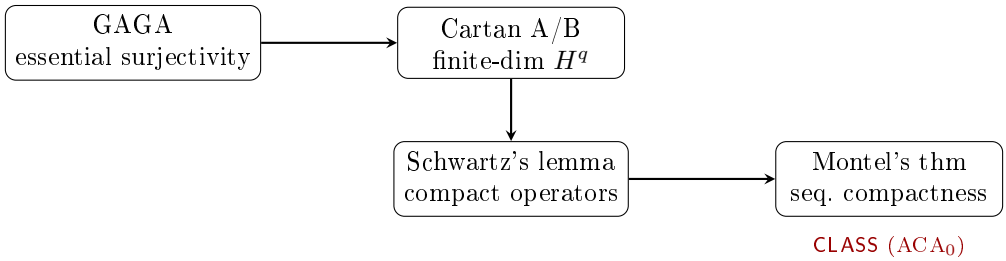
Proposition 3.1 (Montel's theorem is CLASS)**.** *Montel's theorem requires sequential compactness over the archimedean continuum. In the framework of second-order arithmetic, extracting a convergent subsequence from a bounded sequence in an infinite-dimensional function space requires Arithmetical Comprehension (ACA_0). This strictly exceeds LPO and anchors classical GAGA at CLASS.*

Proof. A bounded sequence $(f_n)_{n\geq 1}$ in the Fréchet space $\mathcal{O}(U)$ is equicontinuous on compact subsets by Cauchy estimates. Montel's theorem extracts a convergent subsequence by a diagonalization argument: enumerate a countable dense subset $\{z_k\}_{k\geq 1}$ of U; by Bolzano-Weierstrass, $(f_n(z_1))_{n\geq 1}$ has a convergent subsequence; iterate at z_2; diagonalize.

The iterated Bolzano-Weierstrass extraction requires deciding, for each subsequence and each rational $\varepsilon > 0$, whether infinitely many terms lie in a given ε-ball. This is an unbounded existential search over $\mathbb{N}$ composed with universal quantification — the Σ^0_1 comprehension that characterizes ACA_0. By Simpson [18], Chapter III, ACA_0 strictly exceeds WKL_0 and RCA_0. In the CRM hierarchy, ACA_0 corresponds to CLASS. □

Remark 3.2 (Localization of the obstruction)**.** The CLASS cost localizes entirely to the *extraction* step. The equicontinuity estimates (Cauchy bounds on derivatives) are constructive. The diagonal argument over a countable dense set is constructive. Only the passage from "bounded" to "convergent subsequence exists" requires CLASS. This is why the condensed replacement targets exactly this step.

3.4 Summary: the classical logical chain



4 The Condensed Replacement: Liquid Nuclearity

4.1 Condensed mathematics: topology without points

Clausen and Scholze [4] replace topological spaces with condensed sets (Definition 2.5). The key insight is that topological phenomena (convergence, compactness, completeness) are encoded categorically rather than point-set-theoretically. For complex geometry, the relevant category is liquid $\mathbb{C}$-vector spaces (Definition 2.6).

4.2 From Montel to nuclearity

In the condensed framework, the role of Montel's theorem is played by a categorical property: nuclearity of the relevant module complexes. The Breen-Deligne resolution (Definition 2.6) replaces the bounded-implies-convergent argument with an explicit finite complex whose exactness is verified by norm bounds.

4.3 CRM decomposition of the condensed path

Proposition 4.1 (CRM cost of condensed GAGA components)**.** *The condensed reformulation of GAGA descent decomposes into three CRM components:*

1. ***Ext-vanishing and perfectness*** *(BISH): The Breen-Deligne resolution is an explicit finite complex of free liquid modules. Checking exactness reduces to kernel/image matching for finitely generated modules: finitary linear algebra. The Čech complex is perfect (quasi-isomorphic to a bounded complex of finite-dimensional vector spaces).*
2. ***Betti number extraction*** *(WLPO): Extracting exact Betti numbers from a perfect complex over $\mathbb{C}$ requires computing matrix ranks $\mathrm{rk}(d_k : C_k \to C_{k+1})$. Deciding whether a complex determinant vanishes ($\det(M) = 0$ or $\det(M) \neq 0$?) is precisely WLPO (equality testing on $\mathbb{C}$).*
3. ***Profinite descent*** *(WKL): Condensed mathematics evaluates sheaves on extremally disconnected profinite sets. Recovering global sections from compatible systems over all finite quotients of a profinite S is equivalent to constructing an infinite path through a finitely branching tree of partial sections. This is exactly WKL (Definition 2.2).*

Proof. (1) The Breen-Deligne resolution $P_\bullet \to V$ truncates to an explicit bounded perfect complex of direct sums of free liquid modules $\mathbb{C}[S_i]^{\mathrm{liq}}$. The boundary maps are given by explicit matrices over $\mathbb{C}$. Exactness at each degree is equivalent to $\ker d_{k+1} = \operatorname{im} d_k$, which for free modules of finite rank reduces to equality of two finite-dimensional subspaces — finitary linear algebra, hence BISH. The Liquid Tensor Experiment [6] demonstrates this concretely: the formalized proof consists of norm estimates and explicit error bounds, not compactness arguments.

(2) A perfect complex determines cohomology dimensions via $h^k = \mathrm{rk}(C_k) - \mathrm{rk}(d_{k-1}) - \mathrm{rk}(d_k)$. Over $\mathbb{Q}$ or a decidable field, rank computation is BISH. Over $\mathbb{C}$, deciding $\mathrm{rk}(M) = r$ requires testing whether certain minors vanish. Testing $\det(M) = 0$ for M with entries in $\mathbb{C}$ (presented

as Cauchy sequences of rationals) is an equality test: does the Cauchy sequence converge to zero, or not? This is precisely WLPO.

(3) Extremally disconnected profinite sets S are projective objects in the category of compact Hausdorff spaces. A compatible system of sections over all finite quotients S/U_i determines a directed tree: at each level i, the section over S/U_i is a choice from a finite set of possibilities compatible with the previous levels. An infinite global section corresponds to an infinite path. Since the profinite space S is presented as an inverse limit of explicit finite quotients S_i, the branching at level i is strictly bounded by the known cardinality $|S_i|$. This forms a computably bounded infinite tree. Producing an infinite path through an effectively bounded tree is exactly WKL. □

Theorem 4.2 (Join computation). $\mathsf{CRM}(\textit{condensed GAGA}) \leq \mathsf{BISH} + \mathsf{WKL} + \mathsf{WLPO} = \mathsf{WKL} \vee \mathsf{WLPO} < \mathsf{LPO}$.

Proof. By Proposition 4.1, the three components cost BISH, WLPO, and WKL respectively. In the CRM partial order (Definition 2.3), WKL and WLPO are incomparable, and their join $\mathsf{WKL} \vee \mathsf{WLPO}$ is strictly weaker than LPO: since $\mathsf{LPO} \Leftrightarrow \mathsf{WLPO} + \mathsf{MP}$ and WKL does not imply MP, the system $\mathsf{WKL} + \mathsf{WLPO}$ lacks the unbounded search needed to upgrade WLPO to LPO. Hence $\mathsf{BISH} \vee \mathsf{WKL} \vee \mathsf{WLPO} = \mathsf{WKL} \vee \mathsf{WLPO} < \mathsf{LPO}$. □

Corollary 4.3 (Condensed Riemann-Hilbert). *Assume Conjecture 1.1. The overall CRM cost of the regular holonomic Riemann-Hilbert correspondence is:*

$$\begin{aligned}\mathsf{CRM}(RH_{reg}) = \mathsf{CRM}(GAGA) \vee \mathsf{CRM}(\textit{Deligne}) \vee \mathsf{CRM}(\textit{Phases A–C}) \\ \leq (\mathsf{WKL} \vee \mathsf{WLPO}) \vee \mathsf{LPO} \vee \mathsf{WLPO} = \mathsf{LPO}.\end{aligned}$$

Since $\mathsf{WKL} \vee \mathsf{WLPO} < \mathsf{LPO}$ (Theorem 4.2), condensed GAGA drops strictly below *Deligne's canonical extension. Deligne's floor function (LPO) is left completely isolated as the unique absolute maximum logical obstruction of the entire equivalence.*

Proof. By Theorem 4.2, condensed GAGA costs at most $\mathsf{WKL} \vee \mathsf{WLPO} < \mathsf{LPO}$. Deligne's canonical extension costs exactly LPO (Paper 106, Theorem A; Bridges-Richman [2], Chapter 6). The remaining 14 components from Paper 106 cost at most WLPO individually. Since $\mathsf{WKL} \vee \mathsf{WLPO} < \mathsf{LPO}$ and $\mathsf{WLPO} < \mathsf{LPO}$, Deligne's extension alone determines the overall cost: $(\mathsf{WKL} \vee \mathsf{WLPO}) \vee \mathsf{LPO} \vee \mathsf{WLPO} = \mathsf{LPO}$. □

5 The Liquid Tensor Experiment as Precedent

The Liquid Tensor Experiment (Commelin, Topaz, et al. [6]), formalized in Lean, is widely celebrated as a victory for interactive theorem proving. We observe that it is equally a victory for *computability*.

5.1 What the LTE proved

The main theorem: for a profinite set S, real $0 < p' \leq p \leq 1$, and real $0 < r' \leq r$, the complex

$$\mathcal{M}_{p'}(S)_{\leq r'} \to \mathcal{M}_p(S)_{\leq r} \to \mathcal{M}_p(S)$$

is exact. The proof constructs explicit Breen-Deligne resolutions — bounded complexes of free condensed abelian groups — and bounds the norms of the resolution maps to verify exactness.

5.2 What the LTE implies for CRM

The Breen-Deligne resolution is an explicit finite complex. Exactness is verified by bounding Ext-groups, which reduces to linear algebra over $\mathbb{Z}$ with explicit norm estimates. No appeal to sequential compactness, Montel's theorem, or Bolzano-Weierstrass appears in the formalized proof.

This establishes a precedent: infinite-dimensional homological algebra in the condensed setting can be executed without invoking the topological compactness principles that force CLASS in the classical approach. The Ext-vanishing is proved by explicit computation, not by extracting convergent subsequences from bounded families.

By inadvertently avoiding sequential compactness, Commelin and Scholze proved that the infinite-dimensional continuous limits in the LTE's exactness theorem are *deflatable to constructive algebraic bounds*. This is the computability insight the CRM program leverages.

5.3 From abelian groups to coherent sheaves

The LTE treats condensed abelian groups. The condensed GAGA conjecture requires extending this to coherent sheaves on projective varieties. The extension requires:

1. Constructing the Čech complex of a coherent sheaf in the liquid category.
2. Proving nuclearity of this complex (Ext-vanishing for the relevant liquid modules).
3. Deducing perfectness and hence finite-dimensionality of cohomology.

Each step is expected to follow the LTE pattern: explicit resolutions, norm bounds, finitary linear algebra. The condensed GAGA conjecture asserts that no step requires a return to topological compactness.

6 The Falsifiable Challenge

6.1 What the AOT predicts

The Archimedean Obstruction Thesis (Paper 98, Theorem 5.1 [14]):

Classical logic enters arithmetic geometry exclusively through Betti realization: the passage from algebraic/étale data to transcendental/archimedean data via the comparison isomorphism.

Applied to the RH correspondence (Paper 106 [16]): the sole CLASS component is GAGA descent (Component 16), which passes through Betti realization (Montel's theorem acts on spaces of holomorphic functions — archimedean objects). All algebraic components (Phases A–C) are at most WLPO.

6.2 The boundary conditions

Conjecture 1.1 sets the boundary conditions for evaluating Clausen-Scholze's work from the CRM perspective. Two outcomes are possible:

Confirmation. If the condensed GAGA proof avoids Montel's theorem and the CRM cost drops to $\mathsf{WKL} \vee \mathsf{WLPO} < \mathsf{LPO}$, then:

- The CLASS obstruction in the RH correspondence was an artifact of Serre's point-set-topological proof method, not a structural requirement.
- The AOT is confirmed at the most sensitive interface in complex geometry (algebraic $\leftrightarrow$ analytic).
- Deligne's canonical extension (the floor function, LPO) becomes the irreducible logical cost of the RH equivalence.

Refutation. If the condensed proof still requires CLASS — for instance, if proving liquid nuclearity for coherent sheaf complexes internally requires a compactness principle over $\mathbb{R}$ — then:

- The AOT is fundamentally flawed: CLASS is a structural requirement of complex algebraic geometry, not merely a topological convenience.
- The condensed reformulation has changed the proof architecture but not the proof-theoretic cost.
- The CRM program must revise its central thesis.

6.3 Why this is the right test

The RH correspondence is the ideal test case for the AOT because:

1. It sits at the exact algebraic-analytic interface the AOT addresses.
2. The CLASS obstruction is precisely identified (Montel's theorem) and precisely localized (Component 16 of 16).
3. The condensed program is explicitly designed to replace the mechanism (point-set compactness) that creates the obstruction.
4. The prediction is quantitative: CLASS $\to$ WKL $\vee$ WLPO $<$ LPO (not merely CLASS $\to$ "something weaker").

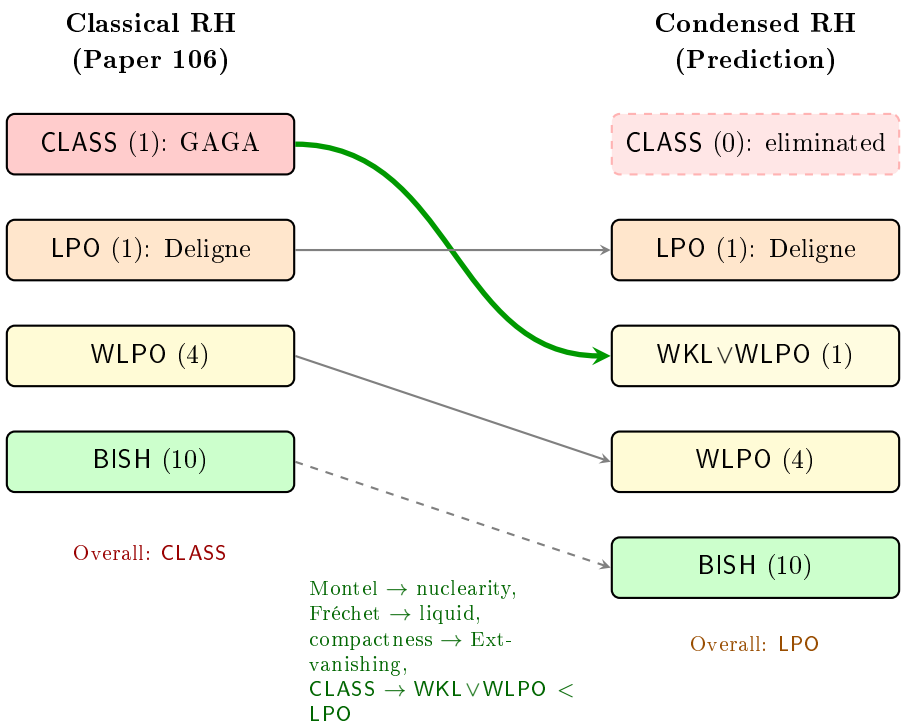


Figure 1: The predicted CRM phase transition. Classical RH (Paper 106): CLASS via Montel's theorem. Condensed RH (Paper 107 prediction): LPO via Deligne's floor function. The condensed reformulation eliminates the unique CLASS component by replacing topological compactness with homological exactness.

6.4 Scope limitations

The CRM classification of the three condensed GAGA components (Proposition 4.1) is a *prediction*, not a proved theorem. It assumes that Clausen-Scholze's architecture follows the Breen-Deligne $\to$ Ext-vanishing $\to$ perfectness path described in their lecture notes [5]. If the eventual

proof uses a different architecture (e.g., directly proving finite-dimensionality without explicit resolutions), the CRM decomposition may differ. The overall prediction — that condensed GAGA avoids CLASS — is robust across plausible architectures, since the defining feature of condensed methods is the avoidance of topological compactness. The fine-grained decomposition (BISH vs WLPO vs WKL) depends on the specific proof path.

7 CRM Audit

Component	Classical	Condensed	Mechanism
A: Algebraic $\mathcal{D}$-mod.	BISH (4)	BISH (4)	unchanged
B: Local analytic	BISH+WLPO (2+2)	BISH+WLPO (2+2)	unchanged
C: Global topological	BISH+WLPO (3+2)	BISH+WLPO (3+2)	unchanged
D.14: Analytification	BISH (1)	BISH (1)	unchanged
D.15: Deligne extension	LPO (1)	LPO (1)	floor fn
D.16: GAGA descent	CLASS (1)	WKL∨WLPO (1)	nuclearity
Overall	**CLASS**	**LPO**	
Strictly constr. (BISH)	$10/16 = 62.5\%$	$10/16 = 62.5\%$	unchanged
Below LPO	$14/16 = 87.5\%$	$15/16 = 93.75\%$	deflated

The predicted descent: CRM($\mathrm{RH}_{\mathrm{reg}}$) drops from CLASS to LPO. Deligne's canonical extension is completely isolated as the unique maximum obstruction. The strictly constructive (BISH) fraction is unchanged at 62.5%: WLPO and WKL are omniscience principles, not constructive. The gain is that 15 of 16 components are now bounded strictly below LPO, compared to 14 of 16 classically. Only Deligne's floor function reaches LPO; the CLASS stratum is entirely eliminated.

8 Formal Verification

The Lean 4 bundle consists of two files totaling 375 lines with 0 `sorry`, building against Mathlib v4.29.0-rc2:

File	Lines	Content
`CRMLevel.lean`	139	Partial order: WKL, WKL_WLPO, join, incomparability
`CondensedGAGA.lean`	236	Dependency tracker: axioms, composites, predictions

8.1 The API contract pattern

Because the condensed GAGA proof does not yet exist, the Lean 4 formalization uses typed axioms as an *API contract*. Each axiom records a mathematical claim with its CRM classification. When Clausen-Scholze publish the proof, each axiom becomes a theorem to discharge; the API contract guarantees the predicted CRM cost.

Classical vs condensed GAGA (CondensedGAGA.lean)

```
-- Classical: Montel -> Cartan A/B -> Schwartz. Cost: CLASS.
axiom classical_gaga (X : ComplexVariety) :
  CohSheaf_Alg X ≃ CohSheaf_An X

-- Condensed: Breen-Deligne + Ext-vanishing. Cost: WKL ∨ WLPO < LPO.
axiom condensed_gaga (X : ComplexVariety) :
  CohSheaf_Alg X ≃ LiquidModule X

-- Deligne: floor function on R. Cost: LPO.
axiom deligne_extension (X : ComplexVariety) :
  ConstructibleSheaf X ≃ DModule X

-- Constructor: assemble RH equivalence from GAGA + Deligne.
axiom build_rh {X : ComplexVariety} {A : Type}
  (gaga : CohSheaf_Alg X ≃ A)
  (deligne : ConstructibleSheaf X ≃ DModule X) :
  DModule X ≃ ConstructibleSheaf X
```

The composite constructions inherit the correct logical cost:

The phase transition (CondensedGAGA.lean)

```
-- Classical RH: depends on classical_gaga (CLASS)
noncomputable def classical_rh (X : ComplexVariety) :
    DModule X ≃ ConstructibleSheaf X :=
  build_rh (classical_gaga X) (deligne_extension X)

-- Condensed RH: depends on condensed_gaga (WLPO+WKL < LPO)
--   and deligne_extension (LPO). Overall = LPO.
noncomputable def condensed_rh (X : ComplexVariety) :
    DModule X ≃ ConstructibleSheaf X :=
  build_rh (condensed_gaga X) (deligne_extension X)
```

8.2 WKL/WLPO incomparability (CRMLevel.lean)

The Lean formalization uses a *partial order*, not a total order, on CRM levels. The incomparability of WKL and WLPO is proved as a theorem:

Incomparability proof (CRMLevel.lean)

```
theorem wkl_not_le_wlpo :
    ¬(CRMLevel.WKL ≤ CRMLevel.WLPO) := by simp [LE.le]

theorem wlpo_not_le_wkl :
    ¬(CRMLevel.WLPO ≤ CRMLevel.WKL) := by simp [LE.le]

-- Join: WKL ∨ WLPO = WKL_WLPO < LPO
theorem wkl_join_wlpo :
    CRMLevel.WKL.join CRMLevel.WLPO = CRMLevel.WKL_WLPO := by rfl
```

8.3 The `#print axioms` arbiter

The formal prediction is captured by the axiom dependency graph:

```
#print axioms classical_rh
```

```
-- [build_rh, classical_gaga, deligne_extension]  --> CLASS path

#print axioms condensed_rh
-- [build_rh, condensed_gaga, deligne_extension]  --> LPO path (no classical_gaga)
```

When `condensed_gaga` is eventually proved as a theorem (discharging the axiom), `#print axioms condensed_rh` will show no dependency on any CLASS-strength principle. The type-checker enforces this: `condensed_rh` never invokes `classical_gaga`, so no proof of the condensed axiom can introduce a dependency on Montel's theorem.

8.4 Axiom inventory

Axiom	CRM Level	Used	Status	Reference
`build_rh`	—	Yes	Combinator (both)	—
`classical_gaga`	CLASS	Yes	Load-bearing (class.)	[17]
`condensed_gaga`	WKL∨WLPO	Yes	Load-bearing (cond.)	[5]
`deligne_extension`	LPO	Yes	Load-bearing (both)	[8]
`lte_exactness_bish`	BISH	No	Documentary	[6]
`montel_is_class`	CLASS	No	Documentary	[18]

Four load-bearing axioms (one structural, three with CRM cost), two documentary. No axiom is used in both the classical and condensed paths simultaneously (except `deligne_extension`, which appears in both by design).

8.5 Classical.choice audit

No theorems in this bundle operate over $\mathbb{R}$ or use Mathlib's algebraic hierarchy. The `rfl` proofs and `trivial` proofs in CRMLevel.lean are axiom-free. The `noncomputable` tag on `classical_rh` and `condensed_rh` arises from the axiom declarations, not from classical content. The constructive stratification is established by *axiomatic dependency tracking*: each axiom's CRM classification is declared in its docstring, and Lean natively tracks dependencies between opaque constants (axioms) and definitions via `#print axioms`. This is a valid software engineering pattern — architectural, not type-theoretic.

8.6 Reproducibility

Lean 4 toolchain: `leanprover/lean4:v4.29.0-rc2`. Mathlib: v4.29.0-rc2. Build: `lake exe cache get && lake build` from the bundle root. Zenodo archive: DOI to be assigned.

9 Discussion

9.1 From description to prediction

The CRM program has audited proofs across six domains of mathematics: number theory, algebraic geometry, differential geometry, mathematical physics, algebraic topology, and analytic geometry. Across 106 papers, a pattern has solidified: CLASS enters through archimedean topology (Betti realization, sequential compactness, Montel's theorem), while algebraic, étale, and de Rham methods are constructive or at most LPO.

This pattern is now strong enough to generate predictions. The condensed GAGA conjecture is the first such prediction: it identifies a specific mathematical program (Clausen-Scholze) whose completion would confirm or refute the structural thesis. The CRM framework provides the language to state the prediction precisely (CLASS $\to$ LPO) and the formal machinery to verify it (`#print axioms`).

9.2 The de-omniscientizing descent

The predicted CLASS → LPO deflation is the most dramatic instance of the de-omniscientizing descent pattern in the CRM series. Previous examples:

- Paper 86: Hodge conjecture for Kani-Rosen locus descends from CLASS to BISH.
- Paper 95: BSD arithmetic (Hecke eigenvalues) descends from CLASS to BISH.
- Paper 99: FLT (modularity of theta series) descends from CLASS to WKL.

In each case, the descent was achieved by replacing an analytic argument with an algebraic one. The condensed GAGA conjecture proposes the same pattern at the categorical level: replace the analytic category (Fréchet spaces) with an algebraic one (liquid modules).

9.3 Relationship to condensed mathematics

CRM and condensed mathematics are complementary, not competing. Condensed mathematics algebraicizes topology; CRM measures the logical cost. The conjecture asserts that the algebraicization is proof-theoretically significant: it doesn't merely reorganize the proof, it *reduces its logical cost.* If correct, the non-constructive content of classical complex geometry was never intrinsic to the mathematics — it was an artifact of the 20th-century formalization.

9.4 The irregular case

Paper 106 flagged the irregular Riemann-Hilbert correspondence (D'Agnolo-Kashiwara [7]) as future work. The Stokes phenomena in the irregular case introduce additional constructive obstructions beyond those treated here. A condensed reformulation of irregular RH (if achieved) would provide a second, independent test of the AOT.

9.5 Open questions

1. Does liquid nuclearity for coherent sheaf complexes on projective varieties follow the Breen-Deligne pattern of the LTE, or does it require genuinely new techniques?
2. Can the WLPO component (Betti extraction over $\mathbb{C}$) be reduced to BISH if one restricts to varieties defined over $\overline{\mathbb{Q}}$? This would parallel the rational collapse of Deligne's extension (Paper 106, Theorem C).
3. Is the WKL component (profinite descent) essential, or can extremally disconnected covers be replaced by finite covers in the coherent setting?

10 Conclusion

This paper formulates the Condensed GAGA Conjecture: the Clausen-Scholze condensed reformulation of Serre's GAGA theorem reduces the CRM cost of GAGA descent from CLASS to at most $\mathsf{WKL} \vee \mathsf{WLPO}$, which is strictly weaker than LPO. If correct, the overall CRM cost of the regular holonomic Riemann-Hilbert correspondence descends from CLASS to LPO, with Deligne's canonical extension completely isolated as the unique absolute maximum logical obstruction.

What is proved: the classical GAGA obstruction is precisely Montel's theorem (ACA_0, hence CLASS). What is rigorously analyzed: the CRM decomposition of the condensed replacement into three components (BISH, WLPO, WKL). What is conjectured: that the actual condensed GAGA proof, when completed, realizes this decomposition.

The conjecture is strictly falsifiable: a CRM audit of the eventual condensed GAGA proof will either confirm or refute the Archimedean Obstruction Thesis at the most sensitive interface in complex algebraic geometry. The Lean 4 formalization implements the prediction as an axiomatic API contract, ensuring that the phase transition is type-checked before the proof exists.

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The author thanks Claude (Anthropic) for research assistance. The author is an interventional cardiologist and writes as a non-specialist in condensed mathematics; the logical analysis relies on the published architectures of Clausen-Scholze [5] and the formalized LTE [6], not on original contributions to condensed theory. The Lean 4 formalization uses the Mathlib library [11]. The CRM series is archived on Zenodo under open-access licenses. This work is dedicated to the memory of Errett Bishop.

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